

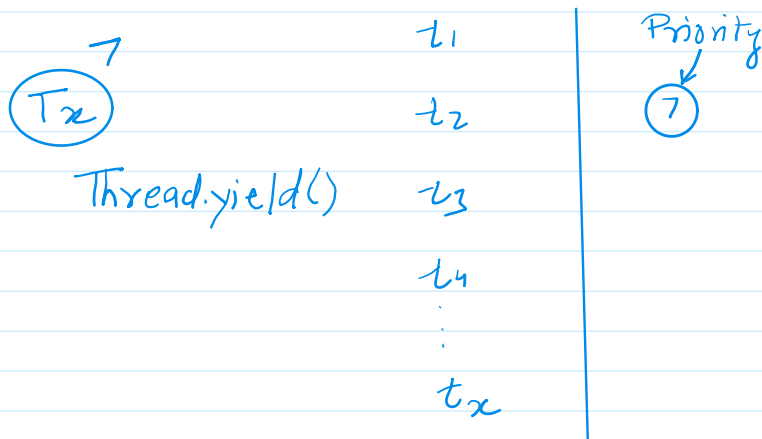
How to prevent thread execution?

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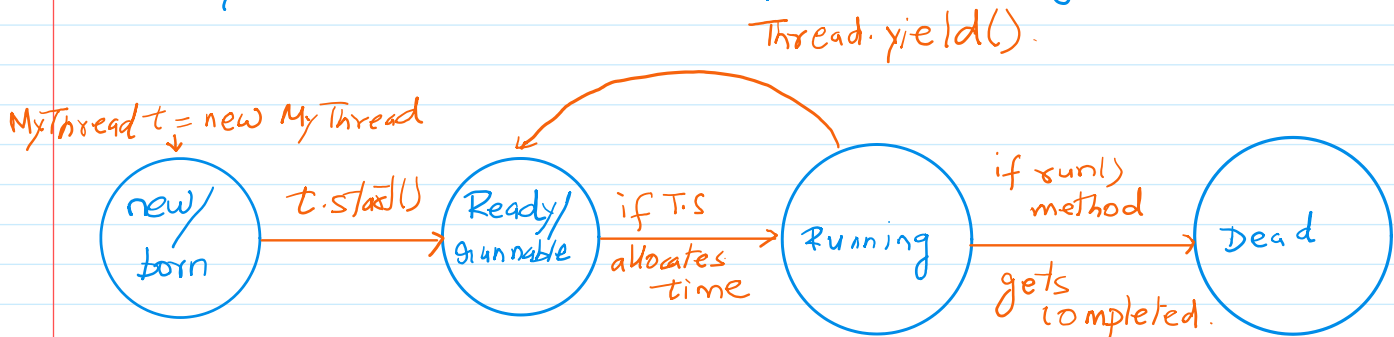
To prevent thread execution we have 3 methods?

- i. `yield()`
- ii. `join()`
- iii. `sleep()`

yield() method: Causes to pause current executing thread to give chance for waiting thread of same priority. If there is no waiting thread or all waiting thread have low priority thread then same thread can continue its execution. If multiple threads are waiting with same priority then which waiting thread will get the chance? We can't expect it depends upon thread scheduler. The thread which is yielded when it will get chance again it depends on thread scheduler and we cannot expect exactly.



1. `yield()` call will pause current executing thread and give chance to other threads of same priority.
2. No waiting thread or low priority thread is existing, then this thread executing is continued.
3. which thread $\Rightarrow$ Thread scheduler (depends on).
4. yielded thread will depend on mercy of Thread scheduler.



package `ThreadState`;

```

public class MyThreadOne extends Thread{
    private int whichChild;
    public MyThreadOne(int x){
        whichChild = x;
    }
    @Override
    public void run() {
        for (int j = 0; j < 10; j++) {
            System.out.println("Child Thread: " + whichChild + " i = " + j);
            Thread.yield();
        }
    }
}

```

```

package ThreadState;
public class MyThreadOneYeildDemo {
    public static void main(String[] args) {
        System.out.println();
        MyThreadOne t1 = new MyThreadOne(1);
        MyThreadOne t2 = new MyThreadOne(2);
        MyThreadOne t3 = new MyThreadOne(3);
        MyThreadOne t4 = new MyThreadOne(4);
        t4.setPriority(Thread.MAX_PRIORITY);
        t1.start();
        t2.start();
        t3.start();
        t4.start();
        for (int i = 0; i < 10; i++) {
            System.out.println("Main Thread");
        }
    }
}

```

Output:

```

Main Thread
Main Thread
Main Thread
Main Thread
Main Thread
Main Thread
Main Thread
Main Thread
Main Thread
Main Thread
Child Thread: 2 i = 0
Child Thread: 3 i = 0
Child Thread: 3 i = 1
Child Thread: 1 i = 0
Child Thread: 4 i = 0
Child Thread: 1 i = 1
Child Thread: 1 i = 2
Child Thread: 3 i = 2
Child Thread: 2 i = 1

```

Note: → pre-emptive scheduling - algorithm

↑
if your o/s supports this algorithm
then it won't provide support yield() method

Child Thread: 3 i = 3
Child Thread: 1 i = 3
Child Thread: 4 i = 1
Child Thread: 1 i = 4
Child Thread: 3 i = 4
Child Thread: 2 i = 2
Child Thread: 3 i = 5
Child Thread: 1 i = 5
Child Thread: 4 i = 2
Child Thread: 1 i = 6
Child Thread: 3 i = 6
Child Thread: 2 i = 3
Child Thread: 3 i = 7
Child Thread: 1 i = 7
Child Thread: 4 i = 3
Child Thread: 1 i = 8
Child Thread: 3 i = 8
Child Thread: 3 i = 9
Child Thread: 2 i = 4
Child Thread: 2 i = 5
Child Thread: 1 i = 9
Child Thread: 4 i = 4
Child Thread: 2 i = 6
Child Thread: 4 i = 5
Child Thread: 2 i = 7
Child Thread: 4 i = 6
Child Thread: 4 i = 7
Child Thread: 4 i = 8
Child Thread: 2 i = 8
Child Thread: 4 i = 9
Child Thread: 2 i = 9